Application of Machine Learning in Predicting Alzheimer's Disease Using Dietary Data

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Introduction

The study focuses on the application of machine learning to predict Alzheimer's disease (AD) by analyzing dietary data. The **main objective** is to explore the relationship between dietary intake and cognitive performance using various machine learning techniques. The dataset used comes from the **National Health and Nutrition Examination Survey** (**NHANES**) for the years 2011-2014 and includes 19,151 observations and 38 variables. The significance of this research lies in the early diagnosis of AD, which is a major global health challenge.

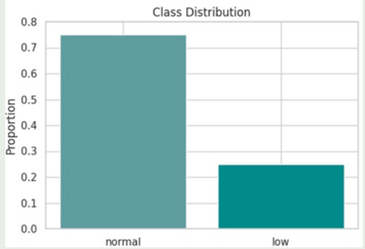
Machine Learning Models Used

The study employed several machine learning models to predict cognitive performance, tested using three cognitive assessments: **CERAD Word List Memory Test**, **Animal Fluency Test**, and **Digit Symbol Substitution Test** (**DSST**). The primary models include **Random Forest** (**RF**), known for its robustness and accuracy, and **XGBoost** (**XGB**), an efficient boosting algorithm with high precision and recall. The **K-Nearest Neighbors** (**KNN**) algorithm, a simpler method, demonstrated moderate performance. The **ensemble of RF and XGB** combined the strengths of both models, enhancing overall predictive capability. Key evaluation metrics included accuracy and AUC (Area Under the Curve).

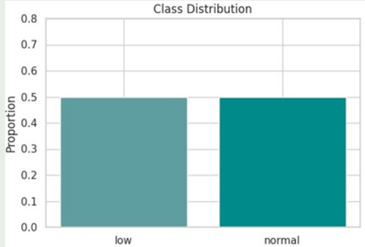
	cerad_score	animalfluency_score	digitsymbol_score
cerad_score	1.000.000	0.383893	0.476524
animalfluency_score	0.383893	1.000.000	0.504925
digitsymbol_score	0.476524	0.504925	1.000.000

Methodology

The study's methodology included various data preprocessing steps. To address class imbalance, the **ROSE algorithm** was applied, balancing the dataset. The data were split into training (80%) and testing (20%) sets. Feature selection was performed to identify the most relevant predictors of cognitive performance. Model evaluation was carried out using metrics such as accuracy, precision, recall, and AUC, ensuring reliable and robust results.



IMBALANCED CLASSES



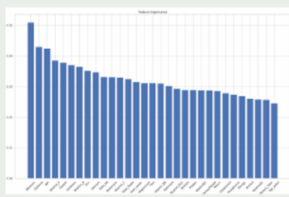
BALANCED CLASSES

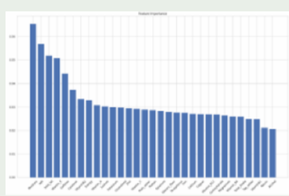
Experimental Results

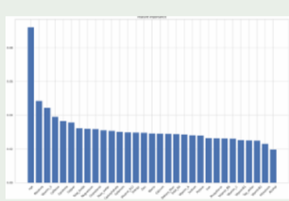
The results showed varying performances among the machine learning models. The stacking model of **RF** and **XGB** achieved **the best performance**, with an accuracy of 0.97 and an AUC of 0.98 in the **CERAD** test. XGBoost performed well in the **DSST**, achieving an accuracy of 0.90 and an AUC of 0.97. **KNN** reported lower performance, with an accuracy of 0.76 and an AUC of 0.82, indicating limitations in class discrimination. Advanced models like RF and XGB proved superior for predicting cognitive performance.

Test	Model	Accuracy	Precision (Low)	Precision (Normal)	Recall (Low)	Recall (Normal)	AUC
CERAD	K-Nearest Neighbors (KNN)	0.76	0.91	0.62	0.70	0.87	0.82
	Random Forest (RF)	0.88	0.93	0.84	0.92	0.92	0.96
	XGBoost (XGB)	0.91	0.98	0.85	0.87	0.97	0.99
	Models Stacking (RF & XGB)	0.97	0.95	0.98	0.98	0.96	0.98
	Ensemble RF and XGB with Prediction Probabilities	0.92	0.97	0.86	0.88	0.97	0.99
Animal Fluency	K-Nearest Neighbors (KNN)	0.76	0.91	0.61	0.91	0.87	0.84
	Random Forest (RF)	0.90	0.94	0.87	0.92	0.92	0.97
	XGBoost (XGB)	0.91	0.95	0.86	0.87	0.97	0.98
	Models Stacking (RF & XGB)	0.96	0.94	0.98	0.98	0.96	0.97
	Ensemble RF and XGB with Prediction Probabilities	0.91	0.95	0.87	0.88	0.97	0.98
DSST	K-Nearest Neighbors (KNN)	0.75	0.88	0.63	0.87	0.83	0.83
	Random Forest (RF)	0.87	0.94	0.80	0.92	0.8	0.95
	XGBoost (XGB)	0.9	0.97	0.83	0.97	0.83	0.97
	Models Stacking (RF & XGB)	0.96	0.94	0.98	0.98	0.98	0.97
	Ensemble RF and XGB with Prediction Probabilities	0.91	0.97	0.84	0.97	0.84	0.98

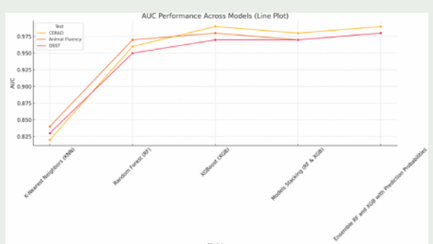
Feature Importance







The analysis of feature importance revealed that variables such as water intake, energy, protein, and total fat had **significant impacts** on cognitive performance. Age, vitamins (e.g., vitamin E and carotene), and minerals were also found to be relevant. These findings highlight **the importance** of hydration and nutrient intake for cognitive health. Analyzing key variables provided valuable insights into how various dietary factors **can influence** cognitive abilities.

Conclusions and Future Directions

The study demonstrated that **ensemble methods**, particularly **the stacking model of RF and XGB**, are **the best approach** for predicting cognitive performance due to their balance of sensitivity and specificity. The integration of ensemble models enhances prediction reliability. **Future research** should include genetic and lifestyle data to further improve predictive capability and the adoption of deep learning techniques to capture complex patterns. Longitudinal studies could deepen the understanding of the long-term effects of dietary patterns on cognitive decline, supporting prevention strategies and personalized interventions.